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## Personal Profile

- Over 10 years' experience analysing large data-sets as a Data Scientist and as a PhD student at CERN.
- Expertise in applying data science techniques, including in a public policy environment.
- Strong communication skills; including presenting data-driven conclusions to business stakeholders.
- My technical skills and experience in fast paced policy environments make me well suited to this role.

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## Technical Skills

- **Computing**      Python (9 years' experience) incl. data science packages; e.g. Pandas, Matplotlib, Scikit-learn. Creating data pipelines using SQL, PySpark and automated code testing (7 years). Bash, Linux and collaborative projects through Git (10 years).
- **Statistics**      Statistical modelling using linear regression and machine learning techniques. Hypothesis testing using A/B and synthetic-control experimentation.
- **Soft Skills**      Presenting analyses; from non-technical colleagues to international scientific conferences. Experience of leadership through line management and mentoring junior data scientists.

## Experience & Education

Mar 2023 -    **HM Treasury (HMT)**  
                 *Principal Data Scientist*

- Delivered and maintain high-profile Financial Market monitoring dashboard.
  - Plotly Dash dashboard using Bloomberg data to summarise market moves every 15 minutes.
  - Fast paced environment; working closely with policy teams to apply updates and fixes, ensuring service for critical end-of-day updates, and enabling monitoring to scale for recent times of high volatility.
  - Dashboard relied on by 100 different users including in Ministerial and Permanent Secretary offices.
- Created new analysis service to project long-term impacts of preventative spending policies.
  - Built regression models using longitudinal survey data to project long-term impacts of policies.
  - Presented and discussed model with academics at LSE, UCL and York to ensure robust methodology.
  - Promoted service to policy teams across Government to gain buy-in and identify impactful policies.
  - Finalised projections for 2 policies, which are being used to inform the current Spending Review.
- Senior leader within team and wider analytic community.
  - Manage 2 apprentices; responsible for their development, project sourcing and oversight of delivery.
  - Proposed and run weekly retro, which is the team forum to celebrate victories and identify pain-points.
  - Gave regular training sessions to other analysts on topics ranging from using Git to Causal Inference.

Nov 2020 -    **Institute for Apprenticeships and Technical Education (IfATE)**

Mar 2023      *Senior Data Scientist*

- Led analysis for T Levels, a new qualification launched in Sept. 2020
  - Built a pipeline using Python, SQL and Pandas to extract, clean and combine data from 3 sources.
  - Developed a T Level Dashboard using Python and Bokeh to enable users to create data insights.
  - Performed bespoke analyses to create data insights to answer specific policy or delivery questions.
- Developed methodology to semi-automate the matching of classification code to each apprenticeship.
  - Used Random Forest model with Word2Vec similarity scores to predict the probability of match
  - Proof-of-concept analysis shows that we can auto-assign 40% of apprenticeships with 88% accuracy.
- Proposed and ran the first ever hackathon at our organisation, encouraging collaboration and innovation.

Jan 2018 -    **Booking.com**

Oct 2020      *Data Scientist - Insights*

- (Jan 2020 - Oct 2020) App Marketing product team, growing the number of app users.
  - Supported product development through analysing and contributing to the design of A/B experiments.
  - Understand and explain the behaviour of the key business metrics to the team.
- (Jan 2018 - Dec 2019) Customer Service Operations Analytics.
  - Determined the key factors driving a long-term deterioration of a performance metric using a linear regression model, as a result the department set-up projects to tackle and mitigate for the key factors.
  - Owned a key performance metric; this involved creating data pipelines, being a visible contact point and helping business stakeholders interpret and use this metric.

Sep 2014 -    **University College London and ATLAS Experiment, CERN**

Dec 2017      *PhD in Particle Physics*

- Applied statistical techniques to search for new Physics in large data-sets from the LHC at CERN.
- Contributed to four public results and presented results at international conferences.

Sep 2010 -    **Merton College, University of Oxford**

Sep 2014      *MPhys Physics*

- Achieved a 2:1 (68%), course involved mathematical and statistical problem solving for a range of situations.